

Five businesses, built out

Product spec, architecture, build plan, validation protocol, pricing, unit economics, go-to-market, battle card, legal posture, metrics and kill criteria — for each of five products.

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How to read the sourcing

Every figure carries where it came from. Nothing is asserted without one of these four marks.

Mark	Means
VENDOR	Read off the company's own page.
REVIEW	Reported by a third-party review or comparison site and not independently confirmed . Several such sites are run by a competitor in the category they price.
STANDARD	A published technical or legal standard.
ASSUMPTION	Mine, marked at every point of use.

Nobody has paid for any of these products. Every commercial figure — churn, sales rate, conversion, what a customer will pay — is an assumption and is marked as one. The accompanying workbook, `Five-Business-Models.xlsx`, is where those assumptions can be changed.

Shared assumptions, and the sizing that reorders the list

Everything in the five buildouts rests on the numbers below. They are here once rather than five times, so that changing one changes all five.

1. The finding that changes the order

The briefs ranked ideas on judgement. Sizing the markets changes the answer, because **break-even is not a number of customers — it is a share of a market**, and these five markets differ by a factor of twenty-one.

Assuming \$6,000/month net personal target, Apple's 15% commission, and 4% monthly churn:

#	Idea	Buyers in US	Price	Break-even customers	Share of market needed
2	Fire & life-safety	19,845	\$89	80	0.40%
8	Behind the wall	127,394	\$49	145	0.11%
6	Floor prep	13,108	\$39	181	1.38%
1	Templating	5,951	\$59	120	2.02%
7	Barrier survey	unknown	\$79	90	cannot be computed

Prices above are the ones the briefs proposed. Idea 1's buildout revises its price from \$59 to \$119 precisely *because* of the 2.02% figure in this table — at \$119 it needs 60 customers, or 1.01%. The workbook uses the revised \$119.

Idea 1 needs eighteen times the market penetration idea 8 does. Two percent of every countertop shop in America is not an impossible number, but it is a different kind of business from one-tenth of one percent of remodelers — it needs near-total category awareness, and it has no room for a niche.

Idea 7's denominator could not be established. California's DSA publishes the CASp list; the page is not reachable from this environment and no published count was found. **That is not a gap in the research, it is the risk itself** — a market you cannot count is a market you cannot forecast, and the buildout says what to do about it before anything is built.

What this does to the ranking

Was	Now	Why
7, 8, 1, 6, 2	8, 2, 7, 6, 1	8 has the largest denominator and the shortest sale. 2 needs the smallest share of its market and has structural retention. 7 is unquantified until you count it. 1 is a good product in a market that may be too small to carry it alone.

The five buildouts are written in your order. **The recommendation at the end is in this one.**

2. Market sizes

Market	Count	Source
Residential remodelers (NAICS 236118)	127,394 businesses / 127,824 establishments (2020 Census)	REVIEW
Fire protection & security system installation contractors	19,845 businesses; \$22.1bn industry, 2026	REVIEW IBISWorld
Flooring contractors (NAICS 238330)	13,108 active companies, ~77,869 employed	REVIEW SICCODE
Countertop manufacturing	5,951 (2024) / 5,967 (2025); \$22.0bn market	REVIEW IBISWorld
Stone fabrication establishments (broader definition)	8,694 with ~100,000 workers (2018)	REVIEW <i>Am. J. Industrial Medicine</i>
Certified Access Specialists (CAsp), California	not found — DSA publishes a list; no count located	—

A note on these counts. Establishment counts include every size of business, and most of these industries are dominated by very small firms — countertop manufacturing is described as "highly fragmented with no companies holding a market share greater than 5%" [REVIEW](#). That fragmentation is good for a flat-priced tool and bad for any strategy that depends on landing a few large accounts.

3. Platform economics

Apple commission	15% under the App Store Small Business Program, for developers with up to \$1M in proceeds in the prior calendar year, and for developers new to the App Store VENDOR .
The threshold is on proceeds, not gross — what Apple paid you, after commission and certain taxes REVIEW .	
Crossing \$1M	Standard commission applies to future sales; you re-qualify the year after proceeds fall back under \$1M REVIEW .
Enrolment	App Store Connect; approval roughly 15 days after fiscal-month end REVIEW .
Marginal cost per customer	\$0. No server, no storage, no per-seat third-party service. This is the single most important number in every model below.

One thing to verify rather than assume. Billing on the web instead of through in-app purchase would replace Apple's 15% with a card processor's ~3%, and on \$89/month that is roughly \$10.70 a month per customer. App Store rules on

external purchase links have been in active legal flux, and **nothing in this research establishes what is currently permitted**. Treat web billing as a question for a lawyer before launch, not a line in the model. Every figure below assumes the full 15%.

4. Churn, and the assumption that decides everything

SMB SaaS monthly churn, industry benchmark	3–5%, with 3–8% also reported REVIEW
Under 3%/month is considered solid for SMB	REVIEW
43% of SMB customer losses occur in the first 90 days	REVIEW

Base case throughout: 4% monthly churn — mid-range, and an [ASSUMPTION](#) in every model. At 4% the average customer lasts 25 months. At 8% they last 12 and half the business is replacing last year's customers.

That 43%-in-90-days figure is the most actionable number in this document. It means **onboarding is not a polish task, it is the retention strategy**, and every buildout below treats first-job success as a product requirement rather than a support function.

5. The standard model

Applied identically to all five:

net per customer per month	= price × 0.85	(Apple's 15%)
lifetime months	= 1 ÷ monthly churn	
LTV	= net per month ÷ monthly churn	
break-even customers	= monthly target ÷ net per month	
new customers needed/month	= existing customers × monthly churn	

Monthly target: \$6,000 net. Change it in the spreadsheet and every figure moves.

6. What is deliberately not modelled

Recorded so a gap is not later mistaken for an oversight:

- **Customer acquisition cost.** No CAC figure appears in these buildouts because none was measured and none was found for these specific trades. Every go-to-market section below is built on **direct, unpaid channels** for exactly that reason — trade associations, distributor relationships, forums, and the phone. Paid acquisition is not modelled and should not be attempted until a real CAC exists.
- **Support cost.** Assumed zero, which is wrong. At 100+ customers it is a real fraction of a person.
- **Any revenue that requires a server.** Hosted links, team seats, and shared archives are named where relevant and excluded from every model.

7. What the model says once you run it

The workbook (`build/Five-Business-Models.xlsx`) projects 24 months at an assumed sales rate. Running it produced one result worth more than the rest of the arithmetic combined.

Steady state = new customers per month ÷ monthly churn. It is the ceiling a given sales rate can ever reach. Beyond it, every new customer is cancelled out by one leaving.

Idea	New/mo ASSUMPTION	Churn	Steady-state ceiling	Break-even	Verdict
2 Fire & life-safety	3	2.5%	120	80	Clears comfortably
1 Templating	3	4%	75	60	Clears
7 Barrier survey	4	4%	100	90	Clears, narrowly
8 Behind the wall	6	4%	150	145	Clears, barely
6 Floor prep	7	4%	175	181	Never breaks even

Floor prep, at seven new customers every month forever, never reaches \$6,000/month. Churn eats it first. It is the only one of the five where the assumed sales rate cannot reach the target at any point in time — and it is the idea whose churn is most likely to be worse than the 4% assumed, because it has the lowest usage frequency.

None of the five reaches break-even inside 24 months at these sales rates. That is not a reason to abandon them; it is a correction to the timeline. Reaching \$6,000/month net is a **two-to-four year** project at these rates, not a twelve-month one. If that is too slow, the dial to move is **price**, not effort: idea 1 needed 120 customers at \$59 and needs 60 at \$119.

Three consequences, and they change what to do:

1. **Floor prep should not be built standalone.** Bundle it — flatness plus moisture plus takeoff, at a higher price and higher frequency — or skip it. The model says the standalone version is arithmetically unable to get there.
2. **Sales rate is the binding constraint everywhere, not product quality.** Every one of these clears break-even eventually except floor prep; what none of them do is clear it *quickly*. Time should go to distribution, not features.
3. **Price is underused across the board.** Every one of these buyers is compared against a four-or-five-figure alternative. These prices were set conservatively and the model says conservatism is expensive.

All of this rests on assumed sales rates. Change the `New customers / month` column on the Dials tab and every one of these conclusions can move.

Buildout — Idea 7: Barrier survey

Working name: PLUMB. Accessibility barrier surveys where every dimension carries proof of how it was obtained.

1. The business in one page

What it is. An iPad and iPhone app that walks an accessibility consultant through a building, measures clearances and slopes, checks each against the governing threshold, and produces two documents off one data set: a barrier report built to be read in litigation, and a priced remediation plan the owner can budget from.

Who pays. The consultant, not the building owner.

The wedge. A barrier survey is evidence. Every dimension in it will eventually be read by somebody with an incentive to attack it. **No existing tool records how a number was obtained** — measured with a tape, read off a sensor, or estimated because the surveyor could not reach. This one marks each, prints the mark, and refuses to certify a barrier finding that rests on an unverified number.

Why the consultant cares more than anyone. Their name is on the report. The provenance record is not a feature they like; it is the file they will want the day somebody questions their work.

Why now. Phone LiDAR plus the device's own inertial sensors can capture clearance and slope together, which is the pair this survey actually needs. That combination did not exist in a consultant's pocket five years ago.

2. Who buys it

Primary ICP. The solo or two-person accessibility consultant who performs barrier surveys as their main business. In California they hold a CASp certification. Outside California they are ADA consultants, accessibility specialists, or architects with an accessibility practice.

Secondary. Facilities and compliance staff at multi-site operators — retail chains, school districts, municipalities — who survey their own portfolio.

Tertiary, and do not chase it first. Architects doing pre-construction accessibility review.

The sizing problem, stated plainly

The number of accessibility consultants in the United States could not be established from public sources.

California's Division of the State Architect publishes a list of certified access specialists; that page was unreachable from this environment, and no published total was found anywhere else. CASp is a California credential and **not a national licence** [REVIEW](#), so outside California there is no register at all.

This is the single biggest unknown in the buildout, and it is not a research gap you can wait out — it is the first task.

What you need	How to get it	Cost
CASp count in California	Download the DSA public list at dgs . ca . gov / casp , or call the CASp Program directly at the number DSA publishes	An afternoon
National consultant count	Count members of the accessibility professional bodies; count firms advertising barrier surveys in the ten largest metros and extrapolate	Two days

Do this before writing a line of code. If the answer is 400 people, this is a lifestyle tool, not a business, and the correct move is to reposition toward the multi-site owner instead.

What is known: BlueDAG charges **\$2,500–\$4,500 per site** for done-for-you surveys [VENDOR](#), and a CASp inspection for a small business averages ~\$1,700 [REVIEW](#). Whoever the buyers are, they bill in four figures per job.

3. Product — v1 scope

The thesis of v1: one surveyor, one site, one report that survives scrutiny.

In scope

Screen	What it does
Site setup	Address, jurisdiction (which standard governs), building type, photos of the frontage
Capture	Walk and scan. LiDAR gives room geometry; the device's motion sensors give running slope and cross-slope; the camera photographs with pose attached
Checkpoint run	The app walks the surveyor through checkpoints by area — parking, path of travel, entrance, restroom, counters. Each has a threshold and a pass/fail
The override	Every sensor value can be replaced by a typed measurement. Typing one re-solves the geometry around it and flips the mark from <code>scanned</code> to <code>measured</code>
The refusal	A barrier finding resting on a <code>scanned</code> value cannot be issued. The app lists what still needs a tape, ranked
Barrier list	Every failure, with its measured value, the threshold it missed, a photo, and its severity
Remediation plan	Each barrier mapped to a fix, priced off the surveyor's own price book, grouped into phases
Report	One PDF: findings, evidence, provenance legend, priced plan

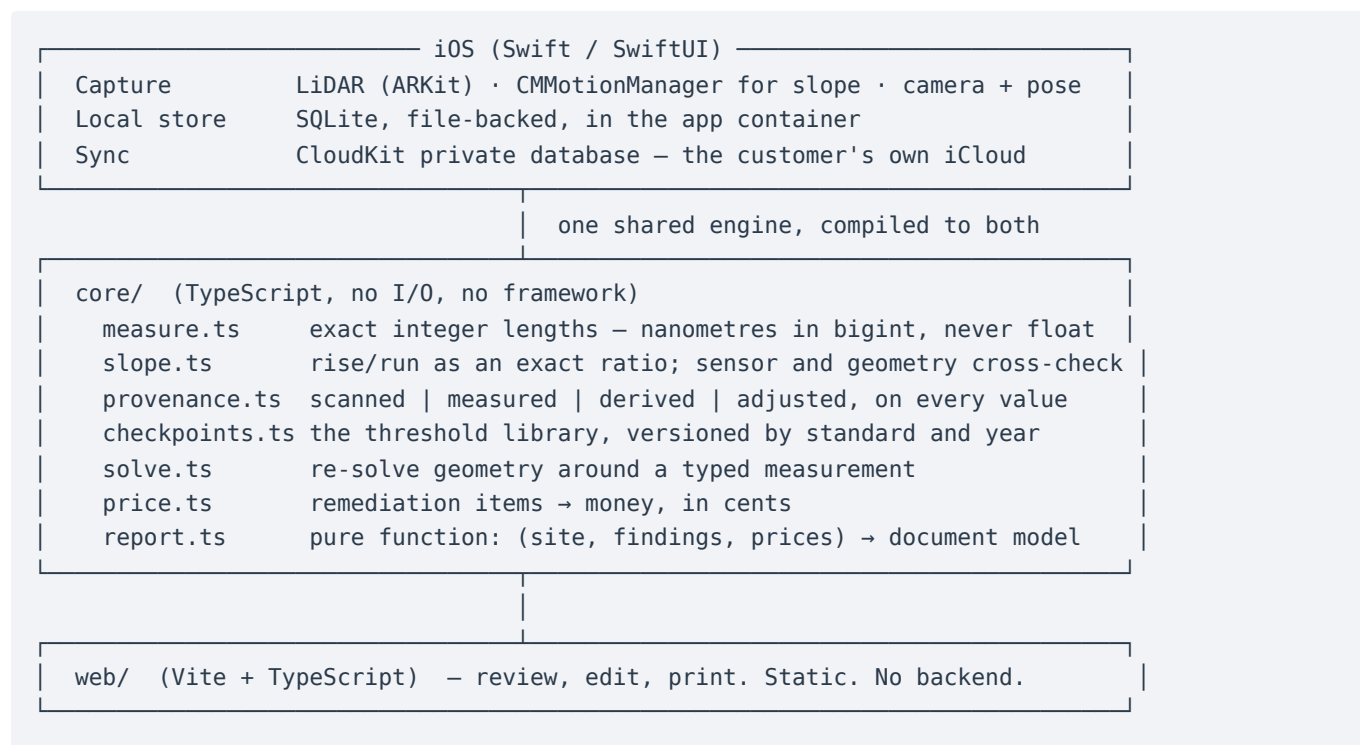
Explicitly out of v1

Recorded so scope does not creep in silently:

- **Every jurisdiction except two.** v1 ships the 2010 ADA Standards and California CBC Chapter 11B. Nothing else.
- **Web/digital accessibility.** A different product with different skills.

- **Multi-surveyor collaboration.** Needs a server.
- **The re-inspection diff.** Ships in v2 and is the second billable event — but it needs v1's data format to be right, which is why the format is designed for it now.
- **Anything that submits or files.** This produces a document. It never becomes the system of record for a legal proceeding.

4. Technical architecture



Five architectural rules, each with a reason:

1. **No float touches a measurement.** Lengths in integer nanometres, slopes as exact rise/run ratios, money in cents. A survey that reports 1:12.0000001 in court is a survey that gets attacked.
2. **The threshold library is data, not code,** versioned by standard and effective date. A survey done in 2027 must be re-printable in 2031 against the thresholds that governed it. **Store the standard version with the survey.**
3. **The engine is shared, not reimplemented.** One `core/` compiled for the phone and the web. Two implementations means two answers.
4. **No server, and it is a legal advantage.** Survey data describes buildings and often names deficiencies before an owner has fixed them. Data you never receive is data you cannot be compelled to produce or lose in a breach.
5. **Slope is measured twice and the disagreement is reported.** Inertial sensors and scan geometry each give a slope. Averaging them hides an error; printing the difference is the honest move and is exactly what an expert witness will look for.

5. Build plan

Estimated in solo weeks. **These are** **ASSUMPTION** **s** — they are ranges because the checkpoint library is the unknown.

Phase	What ships	Weeks
0. Count the market	The denominator. Not code	0.5
1. The engine	core/ — measurement, slope, provenance, solver. Tested to the standard of the numbers it produces	3–4
2. One checkpoint, end to end	Restroom clear floor space: capture → threshold → override → refusal → PDF. Proves the whole spine	2–3
3. The checkpoint library	Parking, path of travel, entrance, restroom, counters — 2010 ADA. The real work	4–6
4. Capture	LiDAR, slope, posed photos, on-device	3–4
5. The report	Findings, evidence, provenance legend, print	2
6. Remediation pricing	Price book, barrier → fix mapping, phased plan	2
7. CBC 11B	California's deltas over the ADA baseline	2–3
8. Field hardening	Ten real surveys, fix what breaks	3
	Total to a sellable v1	21–29 weeks

The critical path is phase 3, and it is not a coding problem. Reading the standard, deciding what each checkpoint measures and how, and encoding the thresholds correctly is research work. Budget it honestly or it will eat the schedule.

6. Validate before you build

Three tests, in order. Each can kill the idea. **Total cost: about a week.**

Test 1 — the denominator. As §2. If California has fewer than ~800 CASps and no plausible national multiple gets you past a few thousand consultants, **stop and reposition to multi-site owners.**

Test 2 — the report. Get one real barrier survey report a consultant has issued. Reproduce it from a paper data model — every field, every finding, every photo. Then ask its author one question: *"would you put your name on this version?"* Their hesitation is the spec.

Test 3 — the slope. Measure ten real ramps with the device's inertial sensors and with a calibrated digital level. **A cross-slope threshold of 1:48 is about 1.19°.** If the phone cannot resolve that reliably, slope must be tape-and-level entry only, and the product is meaningfully weaker. **No figure for phone inclinometer accuracy at this tolerance was found or measured.**

7. Pricing and packaging

Tier	Price	What
Free, forever	\$0	Measure clearances and slopes against thresholds, see pass/fail, one site kept. The whole measuring tool
Professional	\$79/mo · \$790/yr	Barrier list, remediation pricing, the report, unlimited sites
Later: Re-inspection	+\$29/mo	The post-remediation diff. v2

Why \$79. The buyer bills **\$1,700–\$4,500 per survey** VENDOR REVIEW. At one survey a month the subscription is under 5% of a single job's revenue. The sale is not a price negotiation; it is one line of arithmetic.

Why the free tier is safe here — and this is the difference from a consumer-adjacent tool. A free measuring app is useless to the buyer, because **the buyer's product is the report**, not the measurement. Free gets them to trust the numbers. Paid is the only way to get anything they can bill for.

8. Unit economics

Base case: **4% monthly churn** ASSUMPTION, Apple 15%, \$6,000/month target.

Price	\$79/mo
Net after Apple's 15%	\$67.15
Average customer lifetime	25 months
LTV	\$1,679
Break-even	90 customers
New customers/month to hold 90	4
Marginal cost per customer	\$0

Sensitivity

Monthly churn		Lifetime	LTV	Customers for \$6k/mo
2%		50 mo	\$3,358	90
4%		25 mo	\$1,679	90
8%		12.5 mo	\$839	90

Break-even is unchanged by churn — the treadmill is. At 2% you replace 2 customers a month; at 8% you replace 7. Same revenue, completely different life.

At 300 customers: \$20,145/month net, ~\$242k/year. **At \$1M in proceeds Apple's commission doubles to 30%** **REVIEW** — that is roughly 1,240 customers, and the model should not be run past it without re-checking the rate.

9. Go-to-market

No paid acquisition. No CAC figure exists for this buyer and inventing one would be worthless.

First 10 — by hand, by phone

- 1. Get the DSA CASp list. It is a public register of exactly your buyer.
- 2. Call thirty. Not email. **Offer to do their next survey with them, free, in person.** You are not selling; you are watching someone work.
- 3. Ship fixes between visits. The first ten customers should each see something they asked for appear.
- 4. **Charge from the first one.** A free pilot teaches you nothing about whether anyone will pay.

First 100

Channel	Why it works here
The CASp register	A published list of your entire California market. Most industries do not have this
CASp continuing education	Certification requires ongoing education. Being useful in that room reaches the buyer with their professional hat on
The report itself	Every survey lands on a building owner's desk with your name in the footer. That is distribution the customer pays <i>you</i> to perform
Accessibility professional bodies	Conferences and member directories
Expert-witness adjacency	Consultants who testify are the ones who care most about provenance. They are also the most credible references

The sales script

"How do you record which measurements you took with a tape and which you eyeballed?"

— *Nobody has a good answer, because no tool asks for one.*

"When somebody challenges a finding two years later, what do you show them?"

— *This is the whole product. Let them arrive at it.*

Never lead with scanning speed. Every scanning app leads with speed, they have all heard it, and it is not what the buyer is afraid of.

10. Battle card

Competitor	Their position	Where they win	Where you win
BlueDAG	The category software; also sells surveys at \$2,500–\$4,500/site VENDOR	Established, Title II focus, institutional relationships	They compete with their own customers by selling surveys. You never do. Their subscription price is unpublished; yours is on the page
Tape, level, clipboard	What most surveys actually use	Free, trusted, no learning curve	Retyping, no provenance record, no priced plan
General scanning apps	Polycam, Scaniverse	Great meshes	No thresholds, no pass/fail, no report, no provenance
Doing nothing	The real competitor	Costs nothing today	Only loses when a demand letter arrives

The line that does the work: *BlueDAG sells surveys. We sell you the tool to sell yours.*

11. Legal and risk

This section is not optional and is worth a lawyer's hour before launch.

Risk	Posture
A missed barrier leads to a suit, and you are named	The provenance record is the defence — it shows exactly what was and was not verified. Design it as a defence from the first commit , not retrofitted
Unauthorised practice	The app never certifies compliance and never issues a legal opinion. It records measurements and compares them to published thresholds. The consultant certifies. The app reports. Say this in the product, not just the EULA
Stale thresholds	Every survey stores the standard version it was run against and prints it. A survey is re-printable years later against the rules that governed it
"CASp report"	A CASp report has statutory meaning in California. Do not use the term unless a California attorney confirms the output qualifies
Insurance	Professional liability / E&O before the first paying customer
Data	Surveys describe buildings and their unfixed deficiencies. No server means no breach surface, and say so in the sales conversation — it is a selling point

12. Metrics and kill criteria

Instrument almost nothing. Four numbers, collected only from customers who opt in:

- 1. **Surveys completed per customer per month** — the usage number that predicts renewal
- 2. **Percentage of dimensions human-verified** — if it is near 100%, the sensor adds nothing; if near 0%, nobody trusts it
- 3. **Free → paid conversion**
- 4. **Month-3 retention** — 43% of SMB losses happen in the first 90 days REVIEW

Kill criteria — decide these now, while it is cheap

Signal	Window	Action
Denominator under ~2,000 nationally	Week 1	Stop. Reposition to multi-site owners
Phone inclinometer cannot resolve 1:48	Week 1	Slope becomes manual entry; re-score the idea
10 surveys shipped, no consultant will pay	Month 4	Stop. The report is not the bottleneck you thought
Month-3 retention under 60%	Month 6	Onboarding, not features. Fix or stop

13. The 90-day plan

Weeks	Focus	Done means
1	Count the market. Slope test. Get one real report	A denominator, a slope accuracy figure, and a spec
2-5	core/ — measurement, slope, provenance, solver	Engine tested to the standard of its own output
6-8	One checkpoint end to end, restroom clear floor space	Capture → override → refusal → PDF, on a phone, in a real restroom
9-12	Checkpoint library, 2010 ADA, five areas	A real survey of a real small business, done entirely in the app

At day 90 you should have walked one real building with one real consultant and produced one report they would sign. That is the milestone. Not a feature count.

Buildout — Idea 8: What's behind this wall

Working name: OPENWALL. The pre-drywall record, sold to the homeowner as a line on the invoice.

The strongest business of the five. Largest market, shortest sale, and the only one where the contractor makes money rather than spends it.

1. The business in one page

What it is. For the two or three days a renovation has its walls open, the contractor walks each room with a phone. LiDAR captures the geometry; the camera photographs continuously with its exact position attached. The app turns that into a permanent, positioned record of every pipe, wire, duct and stud — and issues it as a self-contained file the homeowner keeps forever.

Who pays. The contractor pays a subscription. **The homeowner pays the contractor \$150–\$400 for the record**

ASSUMPTION.

The wedge — and it is a business-model wedge, not a technical one. Every competitor sells a contractor a subscription to store photos. This sells the contractor **a product to resell at 100% margin on a job where the walls were already open**. Nobody in the category has noticed that the person who benefits most is not the contractor at all.

Why the contractor says yes. It is a differentiator on the bid, a margin line on the invoice, and twenty minutes of work. The pitch to the homeowner writes itself: *"When you sell this house, or the next guy opens this wall, you'll have this."*

Why it is defensible. Not the computer vision — that is copyable. **The artefact.** A record format that homeowners own, that transfers with the property, and that accumulates in the housing stock. Every record issued is a homeowner who expects one on their next job, and asks the next contractor for it.

2. Who buys it

Primary ICP. The residential remodeler doing whole-room or whole-house work — kitchens, baths, additions — where walls come open. Two to fifteen people. **127,394 businesses in the US** **REVIEW**.

Secondary. Electricians and plumbers, who could sell the record on their own rough-in.

Tertiary. Custom home builders, where the record is a closing document.

The end beneficiary is the homeowner, and they are the one you eventually market to — but they are not the buyer, and confusing the two kills the plan.

Why the denominator matters here more than anywhere

Residential remodelers	127,394 REVIEW

Break-even customers at \$49/mo	145
Share of market required	0.11%

One-tenth of one percent. Compare idea 1, which needs 2.02%. This is the difference between needing category dominance and needing a niche.

3. Product — v1 scope

The thesis of v1: one room, walls open, twenty minutes, one file the homeowner can open in ten years.

In scope

Screen	What it does
Job setup	Address, homeowner name, rooms
Capture	Walk the open room. LiDAR builds the wall geometry; the camera shoots continuously with pose attached
Trace	Tap a pipe or cable in one frame, tap it again in another from a different angle. The app triangulates a real 3D position. Label it: cold water, 12-2 romex, 3" waste, HVAC supply
Provenance	A run seen and triangulated is <code>measured</code> . A run inferred between two visible points is <code>derived</code> and prints differently
Verify	A tape on one wall per axis anchors the room, same as any measured space
Issue	One self-contained file — geometry, positioned runs, photographs, the provenance legend — that opens in any browser with no app and no login
Invoice	The record becomes a priced line item

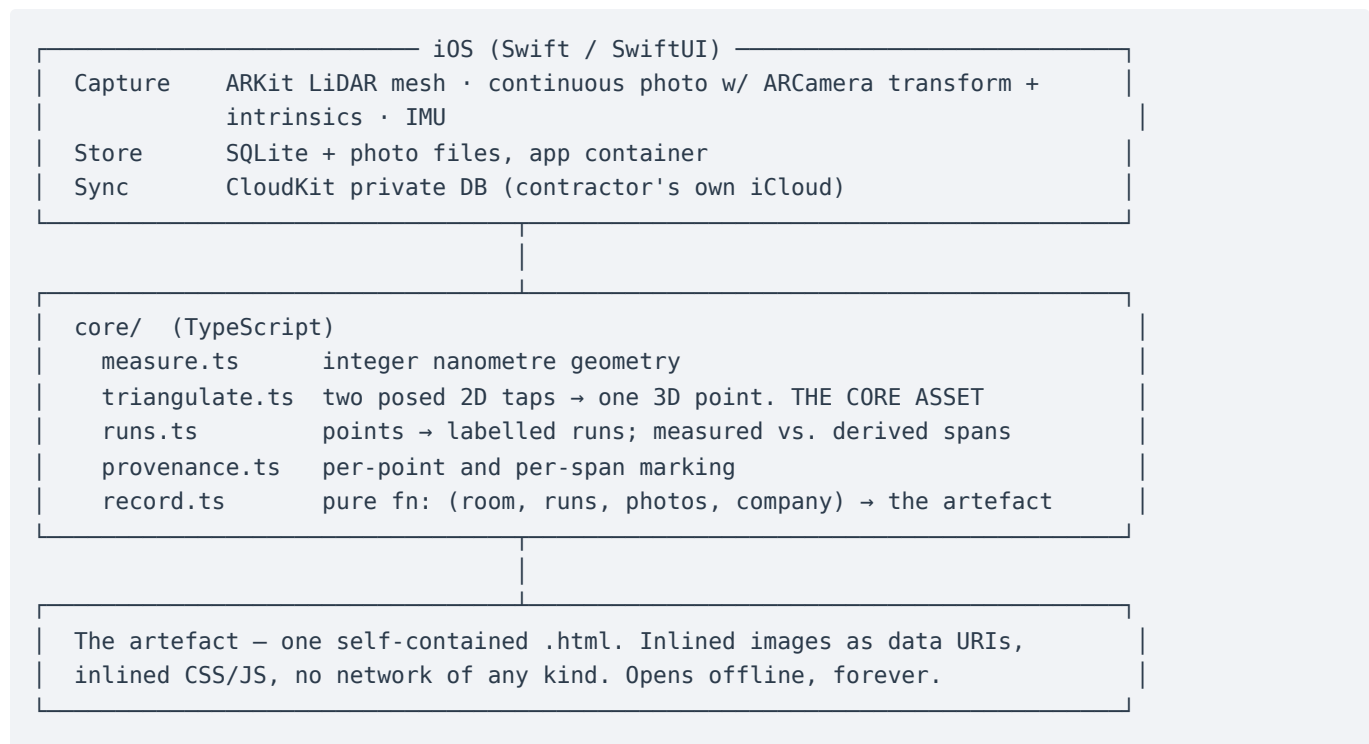
Explicitly out of v1

- **Automatic pipe recognition.** v1 is tap-to-triangulate. A human is standing there and a wrong automatic answer is worse than no answer. *(See §6 — this is the single most important scope decision in the buildout.)*
- **A hosted vault.** Needs a server. The homeowner owns the file. Named as a later tier, priced separately, not in the model.
- **Property transfer / title integration.** Later, and probably a partnership.
- **Whole-house at once.** v1 does rooms.

The one thing v1 must get right

The file has to still open in 2036. That is the whole promise. Self-contained HTML with everything inlined — no CDN, no font host, no external image, no runtime dependency. **A record that needs a server in ten years is not a record.**

4. Technical architecture



The triangulation, precisely. Two camera poses are known from ARKit. The user taps the same pipe in each. Two rays are cast into world space; their closest approach is the point. **The distance between the two rays at closest approach is the error, and it is reported rather than hidden** — a wide miss means the taps disagreed and the app asks for a third.

That reported miss distance is the whole trust model. It is also the thing that makes an automatic recogniser unnecessary in v1.

Why no server, restated as a product requirement: the artefact must outlive your company. If OPENWALL shuts down in 2030, every record ever issued still opens. Say that in the sales conversation — a contractor who has been burned by a dead SaaS understands it immediately.

5. Build plan

Phase	What ships	Weeks
0. Five phone calls	The demand test. Not code	0.25
1. Capture + pose	LiDAR room, continuous posed photos, on device	3
2. Triangulation	Two taps → a 3D point with a reported error. Prototype this before anything else	2–3
3. Runs and labels	Points into labelled runs; measured vs. derived spans	2
4. The artefact	Self-contained HTML, fully inlined, opens offline	3

Phase	What ships	Weeks
5. Room anchoring	Tape on one wall per axis; the room re-solves	2
6. Issue and invoice	Branding, the priced line item	1.5
7. Field hardening	Ten real open-wall rooms	3
	Total to sellable v1	16.75–17.75 weeks

Fastest of the five to a sellable product, because there is no standards library to encode — the thing being recorded is just geometry.

6. Validate before you build

Test 1 — five phone calls. Do this today.

"If it took twenty minutes with the walls already open, would you put \$250 on the invoice for a permanent record of everything behind the drywall?"

You are testing whether they will sell it, not whether they would use it. Every contractor will say they would use a free useful thing. The question is whether they will stand in a kitchen and ask a homeowner for \$250.

- **Four or five yes → build it.**
- **Two or three → the price is wrong, or it needs to be bundled. Ask what they'd charge.**
- **Zero or one → stop.** The latent-value problem is real and no product fixes it.

Test 2 — triangulate one pipe. A weekend prototype: two posed taps, one 3D point, printed miss distance. Then put a tape on the real pipe. **Is it within an inch?** An inch is the useful threshold — good enough to find a run, never good enough to cut blind, which the record says explicitly.

Test 3 — the ten-year file. Build the self-contained artefact and open it in Safari, Chrome, Firefox, and on an Android phone, **with the network turned off.** If anything fails to render, the promise is broken.

7. Pricing and packaging

Tier	Price	What
Free, forever	\$0	Capture, trace, view. One record kept. The whole tool
Pro	\$49/mo · \$490/yr	Issue records, unlimited, own branding, the invoice line
Later: Vault	+\$29/mo	Hosted, re-sendable. Needs a server. Priced separately, never absorbed

The contractor charges \$150–\$400 per record [assumption — untested, and Test 1 is what tests it].

The arithmetic that closes the sale: one record a month at \$250 pays for five months of subscription. **Two records pays for the year.** Everything after is margin on work already being done.

This is the only one of the five where the pitch is "this makes you money." Lead with it and never with features.

8. Unit economics

Price	\$49/mo
Net after Apple's 15%	\$41.65
Lifetime at 4% churn	25 months
LTV	\$1,041
Break-even	145 customers
Share of 127,394 remodelers	0.11%
New customers/month to hold 145	6

Sensitivity — churn matters here more than price

Monthly churn	Lifetime	LTV	Replace/month at 145
2%	50 mo	\$2,083	3
4%	25 mo	\$1,041	6
8%	12.5 mo	\$521	12

The retention mechanism is unusual and worth naming. A contractor who has sold three records has three homeowners who expect one. The habit is reinforced by his own customers, not by your app. **That should make churn lower than the 4% base case — but it is unproven, and the model does not assume it.**

At 500 customers: \$20,825/month net, ~\$250k/year, on 0.39% of the market.

9. Go-to-market

First 10

1. **Your own network first.** Remodelers you can stand next to in a room with the walls open. Do the first record with them.
2. **Do one for free — for the homeowner, not the contractor.** Then watch the homeowner's reaction. That reaction is the entire sales asset, and if it is a shrug you have learned the most important thing early.

3. **Film it.** Twenty minutes compressed to ninety seconds, ending on the file opening in a browser. That video is the ad, the demo, and the pitch.

First 100

Channel	Why
The artefact itself	Every record lands with a homeowner, footer-branded. Homeowners ask their <i>next</i> contractor for one. This is the only one of the five with an organic loop
Home inspectors	They open walls, they meet homeowners at the moment they care, and they are natural referrers
Remodeler forums and local NARI/NAHB chapters	Where this buyer talks about winning bids
Real estate agents	A record is a listing feature. Long game, no cost
Electricians and plumbers	Sell the record on their own rough-in

The sales script

"What happens when someone needs to cut into a wall you closed up three years ago?"

— *Let them describe the guessing.*

"What if you'd handed the homeowner a file, and charged them \$250 for it?"

Then show the ninety-second video. **Do not demo the app.** Demo the artefact.

10. Battle card

Competitor	Position	They win	You win
ScanManifold	\$16/user/mo photo, \$24 with 3D measure REVIEW	Cheap, shipping, no seat minimums	Photos in a timeline, not positions in a wall. It is a contractor cost; yours is contractor revenue
DocuSketch	360° capture + scope, restoration-focused, hardware VENDOR	Fast walkthroughs, established in restoration	Aimed at restoration scoping, not a permanent record. Needs their hardware
Procore / Fieldwire / Raken	General construction documentation	Enterprise depth	Per-seat cost to a two-man shop, and no homeowner artefact
The camera roll	What actually happens	Free	Unfindable in four years, no scale, no position

The line: *Everyone else sells you storage. We sell you something to sell.*

11. Legal and risk

Risk	Posture
Somebody cuts into a wall trusting the record and hits a line	The most serious risk in this buildout. The record states on its face that it is a record, not a locator, and never replaces scanning the wall before cutting. <code>derived</code> spans are visually distinct from <code>measured</code> points. The reported triangulation error prints
Contractor liability for what the record shows	The record is dated and describes conditions at that moment. Later trades change things. Say so on the artefact
Homeowner privacy	The file contains the interior of somebody's home. The homeowner owns it; no copy reaches you. Zero breach surface by construction
A record outliving the company	Solved architecturally — self-contained, no runtime dependency. This is a promise, so it must be tested every release

12. Metrics and kill criteria

1. **Records issued per customer per month** — the only number that matters
2. **Percentage of contractors who issue a second record** — habit formed or not
3. **Triangulation miss distance, median** — the trust metric
4. **Month-3 retention**

Signal	Window	Action
Fewer than 4/5 say yes to the \$250 question	Week 1	Stop. Cheapest kill available
Triangulation cannot hit ~1 inch	Week 4	Re-scope to photo-with-position only; re-score
Contractors use it but never charge for it	Month 4	The model is wrong. Reprice as a cost-saving tool at a lower price, or stop
Under 40% issue a second record	Month 6	Novelty, not habit. Stop

13. The 90-day plan

Weeks	Focus	Done means
1	Five phone calls. Triangulation weekend prototype	A go/no-go and a measured miss distance
2–4	Capture: LiDAR room + continuous posed photos	Walk a real open-wall room, get geometry and posed frames
5–7	Triangulation and runs in <code>core/</code> , with reported error	Tap a pipe twice, get a position within an inch of a tape

Weeks	Focus	Done means
8–10	The artefact — self-contained, offline, cross-browser	Opens on four browsers with the network off
11–12	Issue, brand, invoice. One real paid record	A homeowner has paid a contractor for a record made in your app

Day 90 milestone: one real homeowner has paid for one real record. Not a beta list. A transaction.

Buildout — Idea 1: Templating

Working name: **TEMPLATE.** Digital templating for fabrication shops, on a phone instead of a \$25,000 rig.

Read §2 first. The sizing work found a problem this idea did not have when it was ranked on judgement alone.

1. The business in one page

What it is. An iPhone/iPad app that captures the shape of a real, crooked opening — a countertop run, a stair tread, a shower pan — hands back the measurements worth a tape, takes the fabricator's typed numbers, re-solves the outline around them, and exports a CNC-ready DXF plus a priced quote.

Who pays. The small fabrication shop that cannot justify \$18,000–\$45,000 of templating hardware.

The wedge. A phone is not accurate enough to template a countertop, **and that is the product.** The scan gets the outline; the human confirms the edges that matter; the app refuses to export to CNC while any cut edge rests on a sensor's guess. Nobody scraps a slab because a tape was wrong — they scrap it because a number nobody checked looked exactly like a number somebody had.

Why now. Phone LiDAR plus a solver that re-closes a polygon around typed measurements can, in principle, do the geometric job. **Whether it can hold the tolerance is unmeasured, and it is the whole risk.**

2. Who buys it — and the problem with the count

Primary ICP. The 2–10 person countertop or stone fabrication shop doing roughly 4–20 kitchens a month, currently templating with cardboard, or subcontracting it.

Countertop manufacturing businesses, US	5,951 (2024) / 5,967 (2025) REVIEW
Stone fabrication establishments, broader definition (2018)	8,694, ~100,000 workers REVIEW
Industry market size	\$22.0bn REVIEW
Structure	"Highly fragmented, no company above 5% share" REVIEW

The problem

Break-even customers at \$59/mo	120
Share of market required	2.02%

Two percent of every countertop shop in America, just to reach \$6,000 a month. Idea 8 needs 0.11%. **This idea needs eighteen times the penetration.**

That is not fatal — 120 shops is a knowable, callable list, and fragmentation means no incumbent blocks you. But it changes the strategy completely:

- **There is no niche to hide in.** You need near-total category awareness.
- **Price is doing too little work.** At \$59 you need 120 shops; at \$119 you need 60, which is 1.0% of the market. **Given the alternative is an \$18,000 capital purchase, \$59 is probably underpriced.** See §7.
- **Adjacent shapes matter.** Stair treads, shower pans, glass, solid surface, splash-backs. The same engine, a wider denominator.

3. Product — v1 scope

Thesis: one countertop run, captured, verified, cut, and it fits.

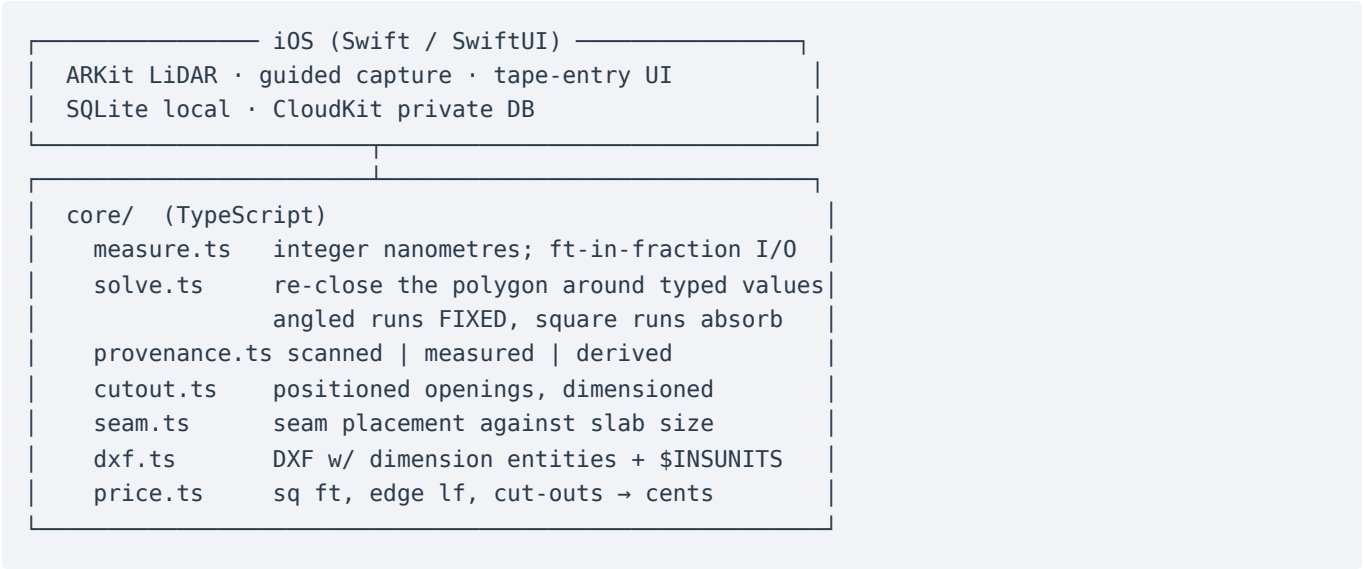
In scope

Screen	What it does
Job setup	Customer, material, edge profile, thickness
Capture	Walk the cabinet run. LiDAR builds the outline; walls, returns, and the front edge
The carry list	The app ranks the measurements worth a tape — back wall length, each return, diagonal — and asks for them, in order
Override + re-solve	Type the real number; the outline re-solves. Angled runs are held fixed while square ones absorb the error, so a chamfer's angle is never quietly stretched
The refusal	Any cut edge carrying a scanned dimension blocks DXF export. Non-negotiable
Cut-outs	Sink, cooktop, faucet holes, from a library, positioned and dimensioned
Seam plan	Where seams fall, against slab size
Export	DXF with dimensions, provenance on every one
Quote	Square footage → price book → edge, cut-out and seam charges → a priced quote

Explicitly out of v1

- **Slab nesting / yield optimisation.** The hardest problem here and not required to prove the concept.
- **Photogrammetry fallback for non-LiDAR phones.** v1 requires a LiDAR device.
- **Direct machine post-processing.** DXF out; the shop's CAM takes it from there.
- **Anything not rectilinear-plus-angles.** Curves are v2.

4. Technical architecture



Three rules:

1. **Integer arithmetic, feet-inches-fractions native.** This trade speaks in sixteenths. A shop that sees 34.9843" loses trust immediately.
2. **The angled-run rule.** A 203 mm chamfer at 70° is ordinary. When arithmetic must give somewhere, it gives in the square runs, never in the angle.
3. **\$INSUNITS must be set correctly in the DXF.** A units header that is wrong or absent is the classic way a CAD export arrives at a machine off by a factor. **Test it against the shop's actual CAM, not a viewer.**

5. Build plan

Phase	What ships	Weeks
0. The slab test	The accuracy answer. See §6	0.5
1. Engine	measure, solve, provenance — the polygon re-solve	3–4
2. Capture + carry list	LiDAR outline, ranked tape list, override	3
3. DXF	Dimensioned, unit-correct, verified in the shop's CAM	2–3
4. Cut-outs and edges	Library, positioning, dimensions	2
5. Seam plan	Slab-size aware	2
6. Quote	Price book, edge/cut-out/seam charges	2
7. Field hardening	Ten real kitchens, cut and fitted	4
	Total to sellable v1	18.5–20.5 weeks

6. Validate before you build

Test 1 — the slab. This is the whole idea and it costs one slab.

Template one real kitchen with a phone plus the human-verified path. Cut it. **Does it fit?**

- Tolerance is roughly **±1.5 mm on a cut edge** [assumption — confirm with a fabricator; manufacturers and shops differ].
- **No accuracy figure for phone LiDAR templating exists in this research and none was measured.**
- Run it three times, three kitchens, before believing the answer.

If it does not fit: stop. There is no version of this that works at any price.

Test 2 — the DXF. Export to a real shop's CAM. Wrong units or missing dimension entities is a silent, expensive failure mode.

Test 3 — the price. Ask ten shops: *"what would you pay monthly to template without buying a rig?"* You are testing whether \$59 is leaving money on the table. Given they are comparing against \$18,000+, expect a higher number.

7. Pricing and packaging — reconsidered

The brief said \$59. **The sizing says that is probably wrong.**

Tier	Price	What
Free, forever	\$0	Template, outline, dimensions, 2 templates kept
Shop	\$119/mo · \$1,190/yr	CNC export, cut-outs, seam plan, priced quote, unlimited

Why \$119, not \$59:

Price	Customers for \$6k/mo	Share of 5,951
\$59	120	2.02%
\$119	60	1.01%
\$199	36	0.60%

The buyer is comparing against **\$18,000–\$45,000 of hardware** [REVIEW](#). At \$119 the app pays for itself against a used LT-55 at \$3,599 [REVIEW](#) in thirty months, and against a new rig in over twelve years. **\$59 signals a toy in a trade where the alternative costs as much as a truck.**

ASSUMPTION — and Test 3 is what tests it.

8. Unit economics

At **\$119/mo**, 4% churn, Apple 15%:

Net after Apple's 15%	\$101.15
Lifetime at 4% churn	25 months
LTV	\$2,529
Break-even	60 customers (1.01% of market)
Replace per month at 60	3

Monthly churn	LTV	Replace/month at 60
2%	\$5,058	2
4%	\$2,529	3
8%	\$1,264	5

At 300 shops (5% of the market — a genuinely hard number): \$30,345/month, ~\$364k/year. The ceiling here is real, and it is set by the denominator, not the product.

9. Go-to-market

First 10

1. **Walk into shops.** This buyer is a physical location with a saw in it. Ten visits in a metro area is two days.
2. **Bring the fitted countertop.** Not a demo — the actual piece that came off a phone template and fitted. That is the only argument that works.
3. **Template one job with them, free, in person.**

First 100

Channel	Why
Stone distributors and slab yards	Every fabricator visits one weekly. The single highest-leverage relationship in this trade
ISFA / trade associations, StonExpo	Where shops evaluate equipment
Fabricator forums and Facebook groups	Highly active, deeply sceptical, and they will test your claims publicly — which is why you must never overclaim accuracy
Used-equipment adjacency	Shops browsing used rigs at \$3,599–\$8,995 REVIEW are shops that have decided they need templating and cannot afford new

The sales script

"How do you template today?" — cardboard, or subbing it out.

"What does a scrapped slab cost you?" — let them say the number.

"This won't let you cut until you've put a tape on the edges that matter."

Never lead with accuracy. You do not have a measured figure, this audience will test it, and being caught overclaiming in a fabricator forum ends the product.

10. Battle card

Competitor	Position	They win	You win
Laser Products LT-2D3D	\$18,000–\$25,000, largest US installed base REVIEW	Proven, trained operators, trusted	Capital cost. A shop without one is not a competitor's customer — they are a non-consumer
Prodim Proliner	\$20,000–\$45,000 REVIEW	Complex and curved work	Price, and curves are v2 anyway
Used LT-55	\$3,599–\$8,995 REVIEW	Cheap entry	Decade-old hardware, no quoting, no provenance
Cardboard	Free	Universal, trusted	Slow, warps, no digital output, no quote
Polycam / Scaniverse	~\$150/yr REVIEW	Great meshes	A mesh is not a template. No closed outline, no tolerance, no DXF a saw accepts

The line: *The rig costs more than the truck you drive to the job. This costs less than the slab you scrapped last month.*

11. Legal and risk

Risk	Posture
A scrapped slab blamed on the app	The refusal is the defence: the app will not export while a cut edge is unverified. Log the verification so the record shows the fabricator typed the number
Overclaiming accuracy	Publish no accuracy figure you have not measured on your own devices, across a real sample. This is both an integrity rule and a legal one
"Confident amateur" risk	A phone makes templating look easy. Templating is a trade. The guided carry list is partly a teaching device — design it that way
Liability cap	EULA reviewed by a lawyer. Consider explicitly disclaiming consequential damages (i.e. the slab)

12. Metrics and kill criteria

- 1. **Templates per shop per month**
- 2. **DXF exports per template** — the conversion from measuring to cutting
- 3. **Percentage of cut edges human-verified** — should be ~100%; if not, the refusal is being worked around
- 4. **Month-3 retention**

Signal	Window	Action
The slab does not fit, three attempts	Week 1	Stop. No recovery.
DXF fails in real CAM	Week 4	Fixable, but fix before selling
Ten shops, none will pay \$119	Month 4	Reprice to \$59 and re-run the market math — or stop
Cannot name 300 reachable shops	Month 6	The denominator is binding. Pivot to adjacent shapes or stop

13. The 90-day plan

Weeks	Focus	Done means
1–2	The slab test, three kitchens. Nothing else	A measured answer on whether this can work at all
3–6	<code>core/</code> — solver, provenance, ft-in-fractions	A polygon that re-closes correctly around typed values
7–9	Capture and the carry list on device	A real cabinet run captured and verified
10–12	DXF, verified in a real shop's CAM	A countertop cut from a phone template, fitted in a real kitchen

Day 90 milestone: a fitted countertop. If week 2 says it cannot fit, you have spent one slab and two weeks instead of five months.

Buildout — Idea 6: Floor prep

Working name: FLATLINE. Find the high and low spots, quantify the leveller, price the prep, win the argument.

This idea was rewritten once already. The original — a phone that produces an ASTM E1155 F-number — is impossible, and §1 says why. What survived is narrower and better.

1. The business in one page

What it is. A flooring installer scans a slab or subfloor before installing. The app produces a relative deviation map, ranks the spots worth a straightedge, takes the real gaps the installer measures, computes how much self-levelling underlayment the floor actually needs, and produces a priced floor-prep change order with the map and photos attached as evidence.

Who pays. The flooring installer or contractor.

The correction, stated up front. ASTM E1155 is the accepted standard for slab flatness **STANDARD**, and automating it requires accuracy on the order of $\pm 0.5\text{ mm over }5\text{ m}$ **REVIEW**. **A phone cannot reach that, and any product claiming an F-number from a phone is wrong.** The app therefore prints a refusal: *this is not an ASTM E1155 test and must not be used as one.*

Why that refusal costs nothing. An installer never needed an F-number. They need to know whether this floor meets the flooring manufacturer's tolerance, and who is paying to fix it.

The real tolerances, which are what the product is actually built against:

Tolerance	Source class
3/16" over 10 feet	REVIEW
1/8" over 6 feet — "most LVP manufacturers"	REVIEW
1/8" over 4 feet	REVIEW

The stakes. Installing over an out-of-tolerance subfloor produces "lippage, hollow spots, popping seams, or full plank failure within 12–24 months, and most manufacturer warranties will be voided" **REVIEW**. It "usually becomes a 'pay twice' situation" **REVIEW**.

So there is a real, recurring, expensive argument on every flooring job — and the installer currently loses it because they have no evidence.

2. Who buys it

Primary ICP. The 1–10 person flooring installation contractor doing LVP, LVT and engineered wood in residential and light commercial.

Flooring contractors (NAICS 238330), US	13,108 active companies , ~77,869 employed REVIEW
Break-even customers at \$39/mo	181
Share of market required	1.38%

Secondary. General contractors and remodelers who sub out flooring and get billed for prep they cannot verify — they are the *other* side of the same argument and may value it more.

Tertiary. Concrete contractors, who want to prove their slab was in tolerance when they left.

That third group is strategically interesting: the same scan settles the argument from both sides, which means the market is bigger than the flooring count alone — but it is also a conflict, and v1 should pick one side and serve it properly.

3. Product — v1 scope

Thesis: scan a floor, know where to put the straightedge, price the prep.

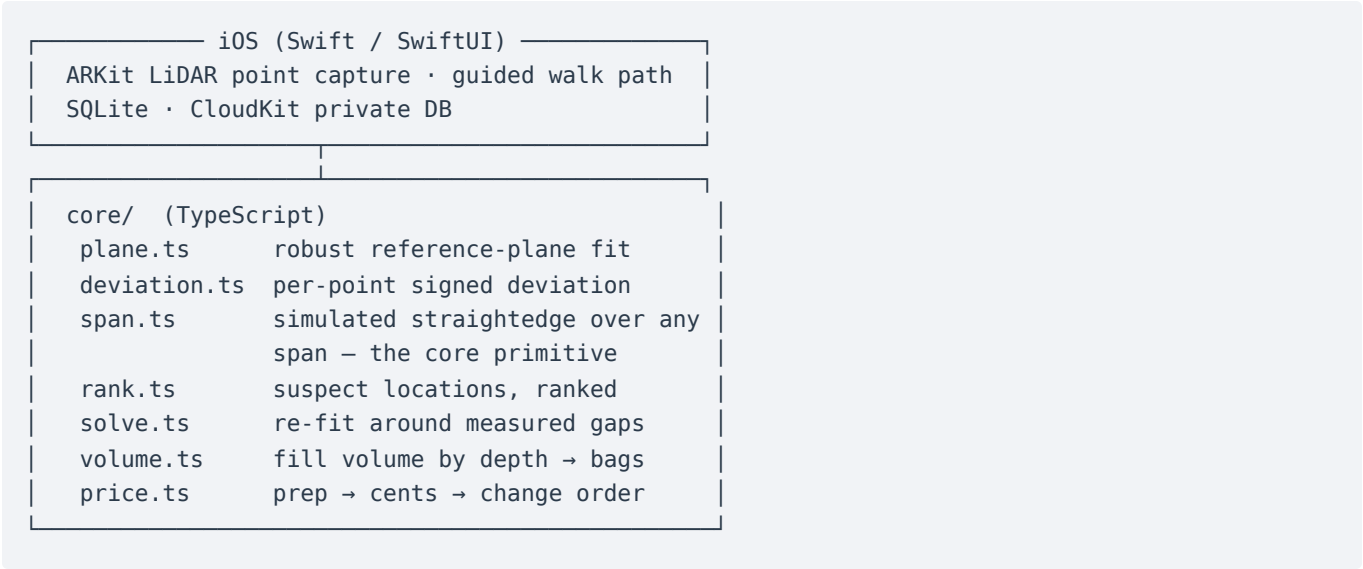
In scope

Screen	What it does
Job setup	Room, flooring product, the tolerance that product requires
Scan	Walk the floor. LiDAR builds a point set; the app fits a reference plane
Deviation map	A heat map of relative highs and lows. Relative, and labelled as such
The straightedge list	Ranked suspect locations — <i>"put a 6-foot straightedge here, and here"</i> — as an on-screen list and a printable card
Measured entry	The installer types each real gap. The map re-solves around the measured points
Verdict	Pass or fail against the product's own tolerance, with the failing spans listed
Leveller quantity	Volume by depth over the failing areas → bags, with waste factor
Change order	Priced prep, with the map, the measured gaps, and photos attached
The refusal	Printed on every output: not an ASTM E1155 test

Explicitly out of v1

- **Any F-number or E1155 claim.** Permanently, not just in v1.
- **Moisture testing.** A different instrument entirely — but it is the natural second product for this buyer.
- **Grinding / high-spot removal quantities.** v1 does fill; v2 does grind.
- **Elevated slabs and deflection.** Slab-on-grade and wood subfloor only.

4. Technical architecture



The one algorithm that matters. `span.ts` simulates laying a straightedge of arbitrary length anywhere on the surface and reports the maximum gap beneath it. That is precisely what the tolerance is written in terms of — *1/8" over 6 feet* — so the product's core primitive should match the specification's own language.

Why relative accuracy might work where absolute does not. Fitting a plane across thousands of points can resolve relative deviation better than any single point's absolute accuracy. **By how much is unknown, unmeasured, and is the whole risk.** See §6.

5. Build plan

Phase	What ships	Weeks
0. The slab test	The accuracy answer	0.5
1. Engine	plane fit, deviation, span simulation, ranking	3
2. Capture	Guided walk, point capture, on-device	2–3
3. Map + straightedge list	Heat map, ranked list, printable card	2
4. Measured entry + re-solve	Type the gap, re-fit	1.5
5. Leveller volume	Depth → volume → bags with waste	1.5
6. Change order	Price book, evidence attachment, PDF	2
7. Field hardening	Ten real floors	2.5
	Total to sellable v1	15–16 weeks

Fastest build of the five. Narrow scope, no standards library, one algorithm.

6. Validate before you build

Test 1 — the ranked list. One afternoon, one slab.

Scan a real slab. Then straightedge it properly, by hand, across a grid.

The question is not whether the app gets the right number. It is whether the app's ranked list contains the spots the straightedge actually found.

- If the top 10 ranked locations include the real problem spots → **the product works**, even if the magnitudes are off, because the human measures the magnitude.
- If the ranking is uncorrelated with reality → **stop**. There is nothing here.

Test 2 — resolution. Can it detect a **3 mm** deviation at all? That is 1/8", the governing tolerance. Place known shims and see. **This figure is unmeasured and unfound.**

Test 3 — will they document it? Ask ten installers: "*if you could prove the slab was out of tolerance before you started, would you use it?*" Listen for hesitation. **Some installers do not want the evidence**, because documenting a bad slab means raising it, and raising it can lose the job. That reluctance is risk #2 in §12 and this test is how you size it.

7. Pricing and packaging

Tier	Price	What
Free, forever	\$0	Scan, deviation map, straightedge list, 2 floors kept
Pro	\$39/mo · \$390/yr	Leveller quantity, priced change order, report, unlimited

Why \$39 and not more. This is a narrow tool used a few times a month. Pricing it like a platform gets it cancelled in month three. **The honest risk is that \$39 is still too much for a tool used twice a month** — see §12.

The bundle question, and it is the important one. A flooring installer would plausibly pay \$79 for *flatness plus moisture plus a materials takeoff*. **This product may want to be the first module of a flooring contractor's tool rather than a standalone subscription**, and that decision should be made before the pricing page is written, not after the first churn cohort.

8. Unit economics

Price	\$39/mo
Net after Apple's 15%	\$33.15
Lifetime at 4% churn	25 months
LTV	\$829

Break-even	181 customers (1.38% of market)
Replace per month at 181	8

Monthly churn	LTV	Replace/month at 181
2%	\$1,658	4
4%	\$829	8
8% (plausible for a narrow tool)	\$414	15

This is the weakest LTV of the five, and the churn risk is the highest, because usage frequency is lowest. Fourteen replacements a month at 8% churn is a treadmill.

The fix is not pricing — it is scope. Adding moisture testing and takeoff raises both frequency and price, and turns a two-uses-a-month tool into a daily one. **Model the bundle before committing to standalone.**

9. Go-to-market

First 10

1. **Flooring distributors and supply houses.** Installers are there weekly, and the counter staff know exactly who has had a callback.
2. **Lead with a lost argument.** Find an installer who ate a floor-prep cost recently. They will tell you the number without being asked.
3. **Scan their next job with them.**

First 100

Channel	Why
Flooring distributors	Highest leverage. They also sell the leveller your app quantifies — an aligned interest
Leveller manufacturers	Your app tells installers how many bags to buy. That is a genuinely aligned partner, and a co-marketing conversation
NWFA / NTCA / flooring trade groups	Where installation standards get argued
Installer Facebook groups and forums	Callback horror stories are the daily content. Your product is the answer to them
The change order itself	Lands on a GC's desk with your branding — and the GC is your secondary ICP

The sales script

"When was the last time you ate a floor-prep cost you didn't bid?"

— Let them say the number. It is usually four figures.

"What if you'd had a map and a measured gap before you started?"

10. Battle card

Competitor	Position	They win	You win
Allen Face D-Meter	\$8,995 REVIEW	The actual standard, court-grade	Price, and installers do not need E1155
Dipstick	Not published; "less than some profilers" VENDOR	The original, trusted	Same
Straightedge + flashlight	\$40, universal	Free, trusted, no learning curve	No map, no record, no quantity, no change order
Laser scan + Geomagic	Enterprise scan-to-heat-map REVIEW	Genuinely accurate	Enterprise cost and skill

The line: *We don't certify your slab. We tell you where to put the straightedge, and what the fix costs.*

Never claim E1155. The people who know the difference are the people whose respect the product needs.

11. Legal and risk

Risk	Posture
Someone uses the output as an E1155 test	The printed refusal is on every export. Repeat it in the app, the report, and the marketing
A floor fails after the app said "pass"	The pass is against measured gaps the installer typed, at the tolerance they selected. Log both. The app never passes a floor on scanned values alone
Leveller quantity is short	Quantities are estimates with a stated waste factor and say so. Never present bag counts as exact
Warranty disputes	The report is evidence of conditions at a moment, not a warranty

12. Metrics and kill criteria

1. **Scans per customer per month** — the frequency number, and the one that decides whether this survives as a standalone product

2. **Change orders generated per scan** — the money conversion
3. **Percentage of scans where a straightedge measurement was actually entered**
4. **Month-3 and month-6 retention**

Signal	Window	Action
Ranked list uncorrelated with real straightedge findings	Week 1	Stop
Cannot resolve 3 mm	Week 2	Stop , or re-scope to "very rough screening" and re-price far lower
Median under 2 scans/customer/month	Month 4	Too narrow. Bundle with moisture + takeoff or stop
Month-6 retention under 50%	Month 6	Confirms the narrow-tool thesis. Bundle or stop

13. The 90-day plan

Weeks	Focus	Done means
1	The slab test and the 3 mm test	A measured answer on whether ranking works
2–4	core/ — plane fit, deviation, span simulation	Simulated straightedge matches a real one on a known surface
5–7	Capture and the deviation map on device	Scan a real floor, get a map an installer recognises
8–10	Straightedge list, measured entry, re-solve	A real floor passed/failed on measured gaps
11–12	Leveller volume and the priced change order	A real installer has issued a real change order from the app

Day 90 milestone: one issued change order. And by then you will also know the scans-per-month number, which decides whether this is a product or a feature.

Buildout — Idea 2: Fire & life-safety inspection

Working name: RISER. The inspection that works in the basement and ends in a signed report and a priced repair.

Best unit economics of the five. Also the hardest build — and \$5 says exactly where the difficulty is.

1. The business in one page

What it is. An iPhone/iPad app for fire-protection contractors. Inspect sprinkler systems, extinguishers, alarms and backflow devices offline in a mechanical room; record what was tested, what failed, and what could not be reached; produce the compliance report and a priced repair quote off the same data, before leaving the building.

Who pays. The 1–15 person fire-protection contractor.

Three wedges, in the order the buyer cares:

- 1. **It works in the basement.** Offline-first is a feature bullet for every competitor and the actual daily reality of this trade.
- 2. **The report says what was *assumed*.** A device that could not be reached, a valve that could not be exercised, a head above a hard ceiling — today these silently become a clean line on a report the inspector signs. This one marks them. **The inspector's signature is on that document;** being able to prove what was and was not verified is professional indemnity, not a feature.
- 3. **The deficiency becomes money the same day.** Found → photographed → priced → quoted → signed, on site. Today that is two or three products and two rounds of retyping, and a meaningful share of deficiencies never get quoted because the retyping never happens.

The pricing finding that is itself the marketing. Neither category leader publishes a price. Inspect Point does not display pricing; BuildingReports lists "Contact for pricing" VENDOR. A search for per-device pricing concluded both "use custom/quote-based pricing models rather than transparent per-device pricing structures" REVIEW. **A published, flat, unlimited number is a competitive act in a market where nobody will say what they charge.**

2. Who buys it

Primary ICP. The small fire-protection contractor doing recurring inspection and service — NFPA 25 sprinkler, NFPA 10 extinguishers, NFPA 72 alarms.

Fire protection & security system installation contractors, US	19,845 businesses REVIEW
Industry size	\$22.1bn (2026) REVIEW , 3.0% CAGR 2020–2025 REVIEW
Break-even customers at \$89/mo	80
Share of market required	0.40%

Note the count is broader than pure fire-sprinkler contractors — it includes security system installers. The true fire-only denominator is smaller and was not separately found. **ASSUMPTION**: **at least half, so ~10,000; break-even is then 0.8% of market, still the second-best of the five.**

Secondary. Facilities departments inspecting their own buildings. **Tertiary.** Authorities having jurisdiction — a different product; do not chase.

3. Product — v1 scope

Thesis: one NFPA 25 annual sprinkler inspection, done offline, accepted by an AHJ.

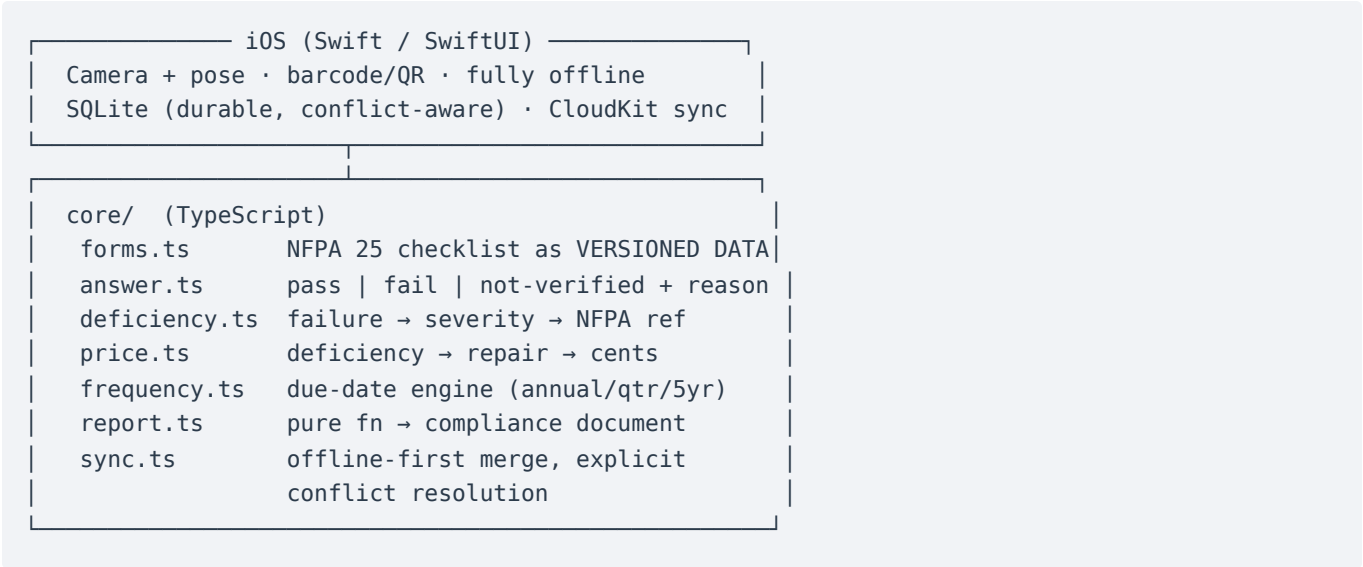
In scope

Screen	What it does
Building + device inventory	Systems, risers, devices, with barcode/QR identity per device
Inspection run	The NFPA 25 annual checklist, walked in order, entirely offline
Three-state answers	Pass / Fail / Not verified — with a reason (inaccessible, hard ceiling, valve seized, tenant refused entry)
Evidence	Photos with camera pose and timestamp, attached to the device
Deficiency	Each failure becomes a record with severity and NFPA reference
Priced repair	Deficiency → repair item → price book → quote, on site
Signature	Building rep signs on the device; audit trail retained
Report	The compliance PDF, with the not-verified items visible rather than absorbed
Frequency tracker	Next due dates — annual, quarterly, five-year — per system
Sync	When signal returns, with real conflict resolution

Explicitly out of v1

- **NFPA 10, 72, and backflow.** v1 is **NFPA 25 annual only**. This is the most important scope decision in the buildout — see §5.
- **AHJ electronic submission.** Every jurisdiction differs. Produce the PDF.
- **Barcode tag supply.** BuildingReports' model puts physical tags on devices. v1 reads existing tags and its own QR codes; it does not build a tagging business.
- **Dispatch and scheduling.** Frequency tracking says *what is due*. It does not become a scheduling product.

4. Technical architecture



Four rules:

1. **The forms are data, versioned by NFPA edition and effective date.** A 2027 inspection must be re-printable in 2032 against the edition that governed it. **Store the edition with the inspection.** Hard-coding a checklist is the single most expensive mistake available here.
2. **"Not verified" is a first-class answer**, never a blank and never silently a pass. This is the entire differentiator and the data model must make it impossible to omit.
3. **Offline is the default path, not a fallback.** Everything works with the radio off; sync is an afterthought that runs later. Building it the other way round always produces an app that fails in a riser closet.
4. **Sync conflicts surface to a human.** Two inspectors, one building, no signal. Silent last-write-wins loses an inspection record — which, on a legal document, is unacceptable.

5. Build plan — and the honest problem

Phase	What ships	Weeks
0. The form test	The go/no-go. See §6	1
1. Offline core	SQLite, sync, conflict resolution. Built first because retrofitting offline never works	3
2. Form engine	Versioned checklist data + renderer + three-state answers	3
3. NFPA 25 annual	The actual checklist content. Research, not coding	4–8
4. Capture	Photos w/ pose, barcode/QR device identity	2
5. Deficiency → quote	Price book, repair mapping, on-site quote	2.5
6. Signature + report	E-signature, audit trail, compliance PDF	2.5

Phase	What ships	Weeks
7. Frequency tracker	Due dates, the retention engine	2
8. AHJ acceptance	Get one real report accepted. Iterate	3
	Total to sellable v1	23–29 weeks

Phase 3 is the risk and it is not a programming risk. NFPA 25 is large, versioned, and applied differently by different AHJs. Encoding one report type correctly is research work. **This is the longest and least certain build of the five**, and it is why the idea scored 2/5 on buildability despite excellent economics.

6. Validate before you build

Test 1 — reproduce one accepted report. One week. This is the go/no-go.

Get one real NFPA 25 annual sprinkler report that an AHJ has already accepted. Reproduce it exactly from a paper data model — every field, every device row, every signature block.

- **If encoding one report type takes a week → buildable. Proceed.**
- **If it takes a month → the other seven report types will take a year. Stop, or narrow to a single jurisdiction and a single system type.**

You will know inside that first week, and it costs nothing but time.

Test 2 — the "not verified" question. Ask five inspectors: *"what do you do today when you can't reach a device?"* If they say "I note it in the comments" or go quiet, the differentiator is real. If they have a clean, confident answer, it is weaker than it looks.

Test 3 — the deficiency-to-quote gap. Ask: *"what share of the deficiencies you find actually get quoted?"* If it is near 100%, the money argument evaporates. If it is 40%, you have found the budget and the headline.

7. Pricing and packaging

Tier	Price	What
Free, forever	\$0	Inspect, photograph, one building's device inventory. The whole inspecting tool
Contractor	\$89/mo · \$890/yr	Reports, deficiency quoting, unlimited buildings and devices, frequency tracking, e-signature

Published on the pricing page, in the largest type on it. In a category where both leaders quote, the number itself is the campaign.

Flat and unlimited, explicitly per-contractor and not per-device. Per-device pricing punishes the contractor who lands a large campus — exactly the customer you want succeeding.

8. Unit economics

Churn assumption is different here, and deliberately so.

The frequency tracker means a contractor's next twelve months of inspection obligations live in the app. Ripping that out mid-year is painful in a way that cancelling a scanning tool is not. **Base case 2.5% monthly** **ASSUMPTION**, **against the 4% used elsewhere.**

Price	\$89/mo
Net after Apple's 15%	\$75.65
Lifetime at 2.5% churn	40 months
LTV	\$3,026
Break-even	80 customers (0.40% of market)
Replace per month at 80	2

Monthly churn		Lifetime	LTV	Replace/mo at 80
2.5% (base)		40 mo	\$3,026	2
4%		25 mo	\$1,891	4
8%		12.5 mo	\$946	7

Highest LTV of the five, by a wide margin, and the lowest replacement treadmill. At 300 contractors: \$22,695/month, ~\$272k/year, on 1.5% of the market.

This is the best business of the five on economics alone. It is fourth on the recommendation only because of phase 3.

9. Go-to-market

First 10

1. **Ride along on ten inspections.** Not demos — carry the clipboard. This trade cannot be understood from a spec.
2. **Find the deficiency-to-quote gap** in each and put a dollar figure on it. That figure is your pitch.
3. **Charge from customer one.**

First 100

Channel	Why
AFSA and NFSA (fire sprinkler contractor associations)	Where this buyer's continuing education and standards discussion happens

Channel	Why
NICET certification communities	Inspectors hold NICET credentials; that is where they gather professionally
Fire-protection distributors	Same weekly-visit leverage as the other trades
The report itself	Every report reaches a building owner and an AHJ with your footer on it
AHJ relationships	An AHJ who finds your report format clear will say so to contractors. The most under-priced channel in this trade
The published price	Write the comparison post nobody else can write, because nobody else publishes a price

The sales script

"What do you do when you can't reach a device?"

"How many of the deficiencies you found last month actually got quoted?"

— That second number is the whole sale. Do not fill the silence.

10. Battle card

Competitor	Position	They win	You win
BuildingReports / ScanSeries	Barcode-scanned device inspection; "Contact for pricing" VENDOR	Real moat: a building already tagged is a building already switched. Deep AHJ familiarity	Opaque pricing, per-application licensing, aimed at larger operations
Inspect Point	Fire inspection & service software; pricing not published VENDOR	Purpose-built, established	No published price. Not offline-first by design
iWorQ	AHJ-side inspection management	Municipal relationships	Different buyer
SafetyCulture	General checklists, published tiers VENDOR	Cheap, flexible, transparent	Generic — no NFPA logic, no deficiency pricing, no compliance report
Paper	What many still use	Free, trusted, works everywhere	Retyped twice; deficiencies never quoted

The line: *Everyone else makes you call for a price. Ours is on the page, it's flat, and it works in the basement.*

On BuildingReports' tag moat, honestly: you will not win a building already tagged. Target contractors *not* on a tag system, and new buildings.

11. Legal and risk

Highest regulatory exposure of the five.

Risk	Posture
An AHJ rejects a report format	Test 1 and phase 8 exist for this. Get one report accepted before selling to anyone
A missed deficiency contributes to a fire	The "not verified" record is the defence. It must be impossible to omit — enforced in the data model, not the UI
Jurisdictions mandating specific systems	Some do. Qualify for it on the first call; do not discover it after onboarding
NFPA copyright	Get advice before shipping. NFPA standards are copyrighted. Encoding <i>what to inspect</i> as your own checklist is different from reproducing standard text, and the line needs a lawyer, not a guess
Stale editions	Store the edition with every inspection; print it on the report
Insurance	E&O before the first paying customer

The NFPA copyright question is a **blocking item**. Resolve it in week one, alongside Test 1.

12. Metrics and kill criteria

1. **Inspections completed per contractor per month**
2. **Deficiencies quoted ÷ deficiencies found** — the money metric and the pitch
3. **"Not verified" answers per inspection** — proves the differentiator is used
4. **Reports accepted by an AHJ without rework**
5. **Month-3 retention**

Signal	Window	Action
One report type takes over a month to encode	Week 4	Stop, or narrow to one jurisdiction and one system type
NFPA copyright advice comes back restrictive	Week 2	Re-scope to a self-authored checklist, or stop
An AHJ rejects the format twice after iteration	Month 6	Format is not the problem — the market is. Stop
Deficiency-to-quote ratio already near 100% in the field	Month 2	The money argument is gone. Re-pitch on offline + provenance alone, at a lower price

13. The 90-day plan

Weeks	Focus	Done means
1	Test 1 (reproduce one accepted report) + NFPA copyright advice	A go/no-go on the two blocking risks
2-4	Offline core: store, sync, conflict resolution	Full function with the radio off; conflicts surface to a human
5-7	Form engine + three-state answers	One NFPA 25 section runs end to end offline
8-11	NFPA 25 annual checklist content	A complete real inspection, run in the app, in a real building
12	Report + one AHJ conversation	An AHJ has looked at a report from your app and said what they think

Day 90 milestone: an AHJ has read one of your reports. Everything downstream depends on that answer, so buy it as early as possible.